GGG-D-296b
February 23, 1968
SUPERSEDING
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May 31, 1957

FEDERAL SPECIFICATION

DIE, THREADCUTTING AND RETHREADING, COLLET, THREADING DIE; AND DIESTOCK (BOLT AND SCREW THREADING, UNIFIED, AMERICAN NATIONAL AND METRIC FORM THREADS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification cover dies producing unified American National and metric form threads, rethreading dies, collets for holding preset adjustable dies and diestocks.

1.1.1 Federal specification coverage. This specification does not include all types of dies, collets, and diestocks indicated by the title of this specification or which are commercially available, but is intended to cover only the types and series which are generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and series. The dies, collets, and diestocks shall be of the following types and series, as specified (see 6.2):

Dies:

Type I. Adjustable, round split dies.

Series A. Unified or American National coarse thread (Symbol NC).
Series B. Unified or American National fine thread (Symbol NF).
Series C. Unified or American National extra fine thread (Symbol NEF).
Series D. Unified or American National special thread (Symbol NS).
Series E. Unified or American National 8-thread (Symbol 8N).
Series F. Unified or American National 12-thread (Symbol 12N).
Series G. Metric thread form.

Type II. Adjustable, rectangular, 2-piece dies with collets.

Series A. Unified or American National coarse thread (Symbol NC).
Series B. Unified or American National fine thread (Symbol NF).
Series D. Unified or American National special thread (Symbol NS).

Type III. Hexagon rethreading dies, solid.

Series A. Unified or American National coarse thread (Symbol NC).
Series B. Unified or American National fine thread (Symbol NF).
Series D. Unified or American National special thread (Symbol NS).

Collets:

Type IV. Collets for type I, 5/8 inch outside diameter dies.

Diestocks:

Type V. Diestocks for type I, round, adjustable, split dies.

Type VI. Diestocks for type II collet dies and type IV collets.

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2. APPLCIABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive, and are not intended to preclude the purchase of dies, collets, and diestocks otherwise conforming to this specification.

3.2 Current product. The dies, collets, and diestocks furnished by the supplier shall be the manufacturer's current product as previously placed in satisfactory use.

3.3 Material. The dies shall be made of water-hardened, oil-hardened, or high-speed tool steel.

3.4 Hardness of dies. The cutting lands of the die threads shall be hardened and tempered in a manner to provide maximum strength and shall show no evidence of partial or total decarburization. Unless otherwise specified herein, the dies shall be hardened throughout to a Rockwell hardness of not less than 60 nor more than 65 on the "C" scale.

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3.5 Marking.

3.5.1 Marking of dies, collets, and diestocks. All dies, collets, and diestocks shall be permanently marked in accordance with MIL-STD-130.

3.5.1.1 Die marking. No distinction shall be made between the marking of the Unified and the American National for dies. Example: A 1/2-13 NC die, whether Unified or American National thread, shall be marked 1/2-13 NC. In either case, American National symbols shall be used. Dies shall be marked with the American National symbol, that is NC, NF, NS. The metric form of thread dies shall be marked with the nominal size and pitch in millimeters, example: 3.5-0.60 mm.

3.5.1.1.1 Left-hand dies. When specified, dies with left-hand thread shall be furnished (see 6.2). Dies with left-hand thread shall be marked "L.H."

3.5.1.2 Diestock marking. In addition to the marking specified in 3.5.1 diestocks shall be marked with the diameter of die or collet accommodated.

3.6 Thread form. The pieces produced by the type I, series A through F, types II and III dies shall comply with the specific thread form, as specified in H28. The threads of the pieces produced by type I, series G, metric dies shall conform to the standards for metric threads.

3.7 Die adjustment. Types I and II dies shall be capable of producing threads having pitch diameters within the limits of the maximum for class 3 fit and the minimum for class 1 fit, as listed in H28.

3.8 Chamfered relief. The chamfered thread crests of dies shall be suitably relieved or otherwise arranged to insure the production of clean, true threads.

3.9 Finish. Threads and surfaces employed in the utility of the dies shall have a smooth finish, free of burrs, nicks, scale, cracks, chatter-marks, or other imperfections which may affect accuracy and long life.

3.10 Type I, adjustable, round, split dies. Type I dies shall be adjustable, round, split dies. Each die shall be made in one piece, having a slot of sufficient width to accommodate the full range of adjustment required. The expansion or contraction of the cutting edges of the dies shall be accomplished, where applicable, by means of a fine-pitched adjusting screw fitted snugly into the screw hole. The outside diameter of the die shall contain two notched or countersink spots midway in the thickness of the die, on a centerline running 90° to the adjusting slot. All sharp edges, other than the cutting edges shall be broken. Each type I die shall be similar to figure 1.

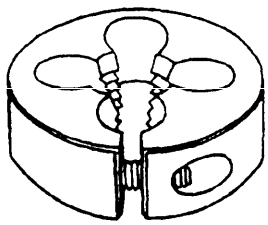


FIGURE 1. Type I, adjustable, round split dies, series A through G.

3.10.1 Series A through F. Series A through F die sizes and dimensions shall conform to table I.

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TABLE I. Type I, series A through F, adjustable, round split dies
Unified or American National Thread series

Nominal size machine screw number	Basic O.D. of thread	Thread symbols and threads per inch				Outside diameter of die 1/	Threads per inch		Outside diameter of die	Thickness of die
		NC UNC	NF UNF	NEF UNEF	NS		NC UNC	NF UNF		
	Inch					Inches			Inches	Inch
0	0.060		80			5/8 or 13/16				1/4
1	.073	64				13/16				1/4
1	.073		72			5/8 or 13/16				1/4
2	.086	56				5/8 or 13/16				1/4
2	.086		64			5/8 or 13/16				1/4
3	.099	48				5/8 or 13/16				1/4
3	.099		56			5/8 or 13/16				1/4
4	.112				36	13/16				1/4
4	.112	40	48			5/8 of 13/16				1/4
5	.125	40				5/8 or 13/16				1/4
5	.125		44			13/16	40	44	13/16	1/4
6	.138	32				5/8 or 13/16				1/4
6	.138	32				1				3/8
6	.138						32		13/16	1/4
6	.138					1	32		1	3/8
6	.138		40			5/8 or 13/16		40	13/16	1/4
8	.164	32				5/8 or 13/16	32		13/16	1/4
8	.164	32				1	32		1	3/8
8	.164		36			13/16		36	13/16	1/4
10	.190	24	32			5/8 or 13/16	24	32	13/16	1/4
10	.190	24	32			1	24	32	1	3/8
12	.216	24				5/8 or 13/16	24		13/16	1/4
12	.216	24				1	24		1	3/8
12	.216		28			13/16		28	13/16	1/4
Nominal size in inches										
1/16	.0625				64	13/16				1/4
3/32	.0938				48	13/16				1/4
1/8	.125				40	13/16				1/4
5/32	.1563				32 and 36	13/16				1/4
3/16	.1875				24 and 32	13/16				1/4
3/16	.1875				24 and 32	1				3/8
7/32	.2187				24	13/16				1/4
1/4	.250	20	28			13/16	20	28	13/16	1/4
1/4	.250	20	28			1	20	28	1	3/8
1/4	.250			32	24	1				3/8
1/4	.250	20	28			1-5/16				7/16
1/4	.250	20	28			1-1/2	20	28	1-1/2	1/2
1/4	.250	20	28			2				5/8
5/16	.3125	18	24			13/16	18	24	13/16	1/4
5/16	.3125	18	24			1	18	24	1	3/8
5/16	.3125	18	24			1-5/16				7/16
5/16	.3125	18	24			1-1/2	18	24	1-1/2	1/2
5/16	.3125	18	24			2				5/8
5/16	.3125			32		1				3/8
3/8	.375	16	24			1	16	24		3/8
3/8	.375	16	24			1-5/16				7/16
3/8	.375	16	24			1-1/2	16	24	1-1/2	1/2
3/8	.375	16	24			2				5/8
7/16	.4375	14	20			1	14	20	1	3/8
7/16	.4375	14	20			1-5/16				7/16
7/16	.4375	14	20			1-1/2	14	20	1-1/2	1/2
7/16	.4375	14	20			2				5/8
1/2	.500	13	20			1-5/16				7/16
1/2	.500	13	20			1-1/2	13	20	1-1/2	1/2
1/2	.500	13	20			2				5/8
9/16	.5625	12	18			1-1/2	12	18	1-1/2	1/2
9/16	.5625	12	18			2				5/8
5/8	.625	11	18			1-1/2	11	18	1-1/2	1/2

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TABLE I. Type I, series A through F, adjustable, round split dies Unified or American National Thread series (con)

Nominal size in inches	Basic O.D. of thread	Thread symbols and threads per inch				Outside diameter of die 1/	Threads per inch		Outside diameter of die	Thickness of die
		NC UNC	NF UNF	NEF UNEF	NS		NC UNC	NF UNF		
	Inches					Inches			Inches	Inch
5/8	0.625	11	18			2	11	18	2	5/8
5/8	.625	11				2-1/2				3/4
11/16	.6875				11 and 16	2				5/8
3/4	.750	10	16			2	10	16	2	5/8
3/4	.750	10	16			2-1/2				3/4
7/8	.875	9	14			2	9	14	2	5/8
7/8	.875	9	14			2-1/2				3/4
1	1.000	8	12		14	2-1/2				3/4
1	1.000	8	12		14	3				1
1-1/8	1.125	7	12			3				1
1-1/4	1.250	7	12			3				1
1-3/8	1.375	6	12			3				1
1-1/2	1.500	6	12			3				1

1/ Tolerances of elements in table I.

Outside diameter of die	Tolerance for outside diameter		Thickness of die	Plus or minus tolerance for thickness
	Plus	Minus		
Inches	Inch	Inch	Inch	Inch
5/8 through 2	-----	1/64	1/4, 3/8, 1/2 and 5/8	1/64
2-1/2 through 3	-----	1/32	3/4 and 1	1/32

3.10.2 Series G. Series G die sizes, dimensions, and tolerances shall conform to table II.

TABLE II. Type I, series G, adjustable, round split dies, metric thread form

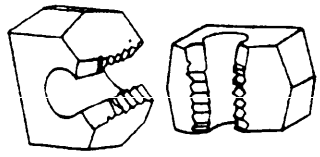
Nominal size	Pitch mm.				Outside diameter of die 1/	Thickness of dies 1/
	French standard	International standard	Spark plug size	Special		
mm					Inches	Inches
1.5	----	----	----	0.35	13/16	1/4
2	0.45	----	----	----	13/16	1/4
2.5	.45	----	----	----	13/16	1/4
3	.60	----	----	----	13/16	1/4
3.5	.60	----	----	----	13/16	1/4
4	.75	----	----	----	13/16	1/4
4.5	.75	----	----	----	13/16	1/4
5	.90	----	----	----	13/16	1/4
5.5	.90	----	----	----	13/16	1/4
6	1.00	----	----	----	13/16	1/4
7	1.00	----	----	----	1-1/2	1/2
8	----	1.25	----	----	1-1/2	1/2
9	----	1.25	----	----	1-1/2	1/2
10	1.50	----	1.00	----	1-1/2	1/2
12	----	1.75		----	1-1/2	1/2
14	2.00	----	1.25	----	1-1/2	1/2
16	2.00	----		----	2	5/8
18	2.50	----	1.50	----	2	5/8
20	2.50	----		----	2	5/8
22	2.50	----		----	2	5/8
24	3.00	----		----	2	5/8

1/ Tolerances of elements in table II.

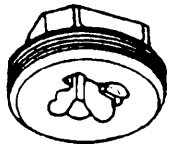
Outside diameter of die	Tolerance for outside diameter		Thickness of die	Plus or minus tolerances for thickness
	Plus	Minus		
Inches	Inch	Inch	Inch	Inch
13/16 through 2	.005	1/64	1/4 and 1/2	1/64
			5/8	1/32

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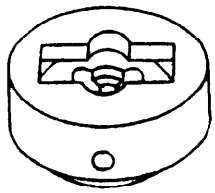
3.11 Type II, adjustable, rectangular, 2-piece dies with collets. Type II dies shall consist of a 2-piece die and a collet composed of a cap and guide. Type II dies shall be similar to figure 2 and shall conform to table III.



Die rectangular (two piece)



Guide



COLLET, COMPLETE

FIGURE 2. Type II, adjustable, rectangular, 2-piece die with collet.

TABLE III. Type II, adjustable, rectangular, 2-piece die with collets

Nominal size machine screw number	Thread symbols and threads per inch			Outside diameter of collet 1/
	NC UNC	NF UNF	NS	
				Inches
2	56	64	32 and 36	1-1/4
3	48	56		1-1/4
4	40	48	32 and 36	1-1/4
5	40	44		1-1/4
6	32	40	36	1-1/4
8	32	36	40	1-1/4
10	24	32	30	1-1/4
12	24	28	32	1-1/4
14			20 and 24	1-1/4
Nominal size in inches				
1/16			64	1-1/4
3/32			48	1-1/4
1/8			40	1-1/4
5/32			32 and 36	1-1/4
3/16			24 and 32	1-1/4 or 2 or 2-3/4
7/32			24 and 32	1-1/4 or 2 or 2-3/4
1/4	20	28	24 and 32	1-1/4 or 2 or 2-3/4
5/16	18	24	32	2 or 2-3/4
3/8	16	24		2 or 2-3/4
7/16	14	20		2 or 2-3/4
1/2	13	20		2 or 2-3/4
9/16	12	18		2-3/4
5/8	11	18		2-3/4 or 4
11/16			11 and 16	2-3/4 or 4

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TABLE III. Type II, adjustable, rectangular, 2-piece die with collets (con.)

Nominal size in inches	Thread symbols and threads per inch			Outside diameter of collet 1/
	NC UNC	NF UNF	NS	
				Inches
3/4	10	16		2-3/4 or 4
7/8	9	14		2-3/4 or 4
1	8	12	14	2-3/4 or 4
1-1/8	7	12		4
1-1/4	7	12		4

1/ Tolerances of outside diameters of collets in table III.

Outside diameter of collet	Tolerances minus
Inches	Inch
1-1/4 or 2 or 2-3/4	1/32
4	3/64

3.11.1 Rectangular 2-piece die. The die shall consist of 2 pieces or sections, and each section shall have 2 lands containing the cutting edges. There shall be adequate chip clearance between the lands. The sides of each section shall have a double bevel for positive reversible positioning in the cap of the collet. The top and bottom die faces shall be parallel and the longitudinal axis of the cutting edges shall be square with the die face.

3.11.1.1 Chamfer. The dies shall be chamfered approximately 2 to 3 threads at the front face and from 1 to 1-1/2 threads at the rear face.

3.11.2 Collet cap and guide.

3.11.2.1 Cap. The cap shall contain a beveled slot which shall act as a snug bearing for the positioning of the die sections. Each cap shall have a screw hole at each end of the die positioning slot. Each screw hole shall be snugly fitted with a fine pitch screw for the adjustment of the dies capable of producing a tight or loose fitting screw thread within the limits of the maximum for class 5 fit and the minimum for class 1 fit, as listed in H-28. All caps shall contain two conical screw spots or grooves 90° from each adjusting screw hole located on the outside diameter to coincide with the location of the die holding screw in type VI diestocks (see 3.14.2). The rear of the cap shall have an internal thread for the reception of the external thread of the guide.

3.11.2.2 Guide. The front end of the screw guide shall be threaded for assembly to the cap. The rear end shall contain a square surface to accommodate a wrench. Each guide shall have a clearance hole to receive and guide the work to be threaded. A different guide will be required for each cutting size of the die and the nominal diameter shall be marked thereon.

3.12 Type III, hexagon rethreading dies, solid. Type III, hexagon rethreading dies shall be of solid construction containing threading lands. The dies shall be designed for dressing over bruised or rusty threads. The clearance between lands shall be of sufficient depth to provide room for the disposal of chips and foreign matter. The dies shall be hexagon in shape and chamfered on both sides. Both faces shall be parallel and square to the axis of the cutting edges. The cutting edge shall have a positive angle and the chamfered thread crests shall be relieved. Type III dies shall be similar to figure 3 and shall conform to the dimensions and tolerances shown in table IV. Threads shall be within the limits of the maximum for class 3 fit and the minimum for class 1 or 1A of thread (whichever is smaller), as listed in H28.

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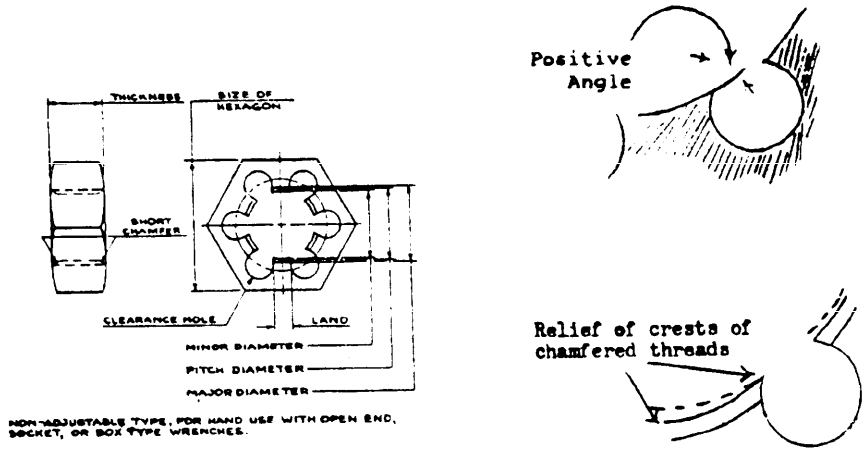


FIGURE 3. Type III, hexagon, rethreading die, solid.

TABLE IV. Type III, series A, B, and D, solid hexagon, rethreading dies

Nominal size in inches	Threads per inch			Dimensions 1/	
	NC	NF	NS	Across flats	Thickness
				Inches	Inch
1/4	20	28		19/32	1/4
5/16	18	24		11/16	5/16
3/8	16	24		25/32	3/8
7/16	14	20		7/8	7/16
1/2	13	20		1-1/16	1/2
9/16	12	18		1-1/16	1/2
5/8	11	18		1-1/4	5/8
11/16	11	16		1-1/4	5/8
3/4	10	16		1-7/16	3/4
7/8	9	14		1-5/8	7/8
1	8	12	14	1-13/16	1
1-1/16				2	1
1-1/8	7	12		2	1
1-1/4	7	12		2-3/16	1
1-3/8	6	12		2-3/8	1
1-1/2	6	12		2-9/16	1

1/ Tolerances for table IV

	Tolerances in inches	
Across flats, sizes 1/4 to 1/2 inch, inclusive	+0.002	-0.010
Across flats, sizes 9/16 to 1 inch, inclusive	.008	.012
Across flats, sizes 1-1/16 to 1-1/2 inch, inclusive	.012	.016
Thickness sizes 1/4 to 9/16 inch, inclusive	±1/32	
Thickness sizes 5/8 to 1 inch, inclusive	±1/16	

3.12.1 Chamfer. The dies shall have approximately one thread chamfered at both the front and rear faces.

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3.13 Type IV, collets for type I, 5/8-inch outside diameter dies. When specified in the contract or order, collets suitable for holding type I, 5/8-inch outside diameter dies shall be furnished for the screw size specified (see 6.2). Collets shall have 7/8-inch (±1/32 inch) outside diameter. They shall be recessed to receive a 5/8-inch outside diameter die, and provided with an integral guide for guiding the work piece centrally into the die. Ample chip clearance shall be provided. Collets shall contain 3 adjusting screws, 2 for compressing and 1 for expanding the die. A conical or oval screw spot shall be provided so that the diestock thumb screw will hold the collet containing the die firmly in the stock. The collets shall be similar to figure 4.

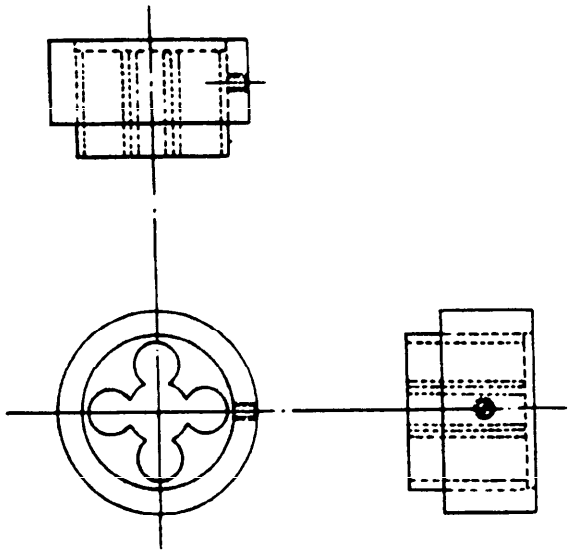


FIGURE 4. Type IV collet for type I, 5/8-inch outside diameter die.

3.14 Diestocks.

3.14.1 General requirements. Diestocks shall have steel or cast iron bodies. Die recesses shall be machined accurately to fit dies and collets of the type and size specified.

3.14.1.1 Handles. The handles shall be of such size and shape as to afford comfortable handgrips and shall be of such strength as to withstand without permanent deformation the torque required to cut maximum capacity threads, when applying the torque on only one end of either of the handle ends. The handles may be either solid or tubular. Handle ends shall be smoothly rounded. The hand gripping portion may be either smoothly finished or knurled.

3.14.2 Type V, diestocks for type I dies. Type V diestocks shall have 3 screw holes placed 90° apart. The centerlines of these holes shall intersect at the center of die recess. These holes shall contain 3 hardened screws. The center screw shall be pointed to a 60° included angle. The other 2 screws shall be rounded at the thread end. The height at the screw hole centerline from the die seat of the die recess shall be slightly less than one half the nominal thickness of the applicable diameter of the dies. The diestocks shall be similar to figure 5 and shall conform to table V.

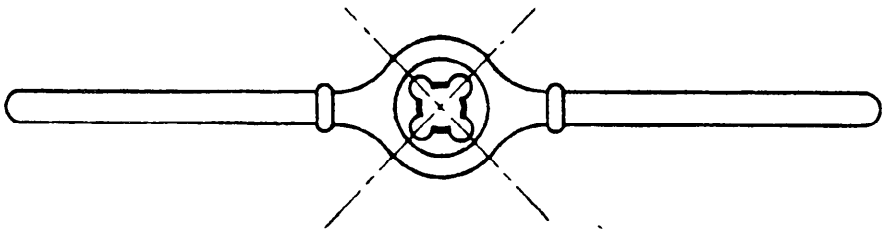


FIGURE 5. Type V, diestock for type I, round adjustable, split dies.

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TABLE V. Type V, diestock for type I, adjustable, round split dies

Diestock sizes	Holding dies, outside diameter	Length overall	
		Minimum	Maximum
	Inches	Inches	Inches
0	5/8	5	6
1	13/16	6	8
1A	1	9	12
2	1-1/2	12	18
2A	2	20	28
3	2-1/2	22	32
4	3	31	43

3.14.3 Type VI diestocks for collet dies. Type VI diestocks shall be for use with type II collet dies and type IV collets. Diestocks of this type shall have a cone pointed, thumb or square head set screw intersecting the collet recess at 90° to the axis of the handles and so positioned as to securely hold the collet in the recess seat. Type VI diestocks shall be similar to figure 6 and conform to requirements of table VI for the diestock size specified (see 6.2).

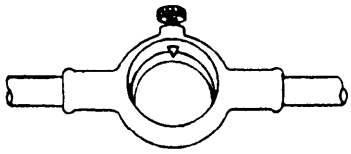


FIGURE 6. Type VI, diestocks for collet dies.

TABLE VI. Type VI, diestock for collet dies

Type of die with collet	Diestock sizes	Diameter of collet recess	Length overall	
			Minimum	Maximum
		Inches	Inches	Inches
5/8 inch dia. round	00	7/8	5	6
Rectangular 2-piece	0	1-1/4	7	10
Rectangular 2-piece	1	2	12	18
Rectangular 2-piece	2	2-3/4	22	32
Rectangular 2-piece	4	4	40	50

3.15 Workmanship. Workmanship of the dies, collets and diestocks shall be of the quality prevailing among manufacturers normally producing dies, collets, and diestocks of the types and classes listed herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All dies, collets, or diestocks of the same size, series, and type presented at one time shall be considered a lot for the purposes of quality conformance inspection.

4.2.2 Sampling for examination. A random sample of dies, collets, or diestocks shall be selected from each lot of material offered for inspection in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be as specified in table VII.

4.2.3 Sampling for hardness and cutting test. A random sample of dies and rethreading dies shall be selected from each lot offered for tests in accordance with MIL-STD-105 at inspection level S4. The AQL shall be as specified in table VII.

TABLE VII. Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	None defined.
1	
Major:	AQL 2.5 percent.
101	Type, size, and thread form of die not as specified.
102	Thread and utility (working) surfaces, not free of burrs, nicks, scale, cracks, or chatter marks.
103	Threads and die threads not chamfered, number of threads chamfered (front and rear faces) not as required chamfered threads not suitable relieved, or not provided with adequate chip clearance.
104	Die not one piece, not adjustable, or adjustment screw not not fine pitched (except type VI).
105	Side of each die section not double beveled (type VI only).
106	Faces of die not parallel, or faces not square with longitudinal axis of cutting edges.
107	Die sections not a snug fit in the cap, die sections not adjustable for tight or loose fit, adjusting screws not snugly fitted in screw holes, screw spots (or grooves) for die-holding screws missing or improperly located.
108	Adjusting screw spots (or grooves) missing or improperly located (except type VI).
109	Guide missing, fails to assemble (screw into) and hold die sections in cap, clearance hole in guide missing or not as required, or rear end of guide not provided with square surface to accommodate a wrench (type VI only).
110	Outside diameter of die not within the specified tolerance (except type VI).
111	Outside diameter of collet not within the specified tolerance (type VI only).
112	Threads not right hand or not left hand, as specified.
113	Adjusting screw thread engagement and smoothness of fit not as specified.
Minor:	AQL 4.6 percent.
201	Thickness of die not within the specified tolerance (except type VI).
202	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.

4.3 Quality conformance inspection.

4.3.1 Examination. Each of the sample dies, collets, or diestocks selected in accordance with 4.4.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table VII. Any dies, collets, or diestocks in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective dies, collets, or diestocks in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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4.4 Test procedures.

4.4.1 Hardness test. Sample dies shall be tested to determine compliance with 3.4. Hardness readings shall be taken on both faces of the die as close to the root diameter as possible without damaging the thread.

4.4.2 Check thread. On types I and II, open the die sufficiently to permit entry with ease of a basin plug thread gage of corresponding nominal size and pitch and with gage in place, close the die until it makes contact with the gage to determine that the die has the correct thread and has sufficient adjustment. On type III, check die thread with a screw pitch gage of the correct number of threads per inch, and die should reject a basic thread plug gage of corresponding nominal size and pitch.

4.4.3 Thread cutting test (types I, II, and III). Two test items shall be drawn from the sample that was selected in accordance with 4.2.1 to be used for thread cutting tests. Each of the two test items shall cut a thread 2 inches long on 10 sample lengths of medium carbon (1035-1050) steel rod, of applicable size. The cut thread, throughout its entire length, shall be within the minimum class 1A and maximum class 3A limits (whichever may be applicable) as listed in H28.

4.4.4 Production check test. In the event of controversy, samples in addition to and exclusive of the number required in a contract or order shall be forwarded to the laboratory designated by the procuring activity for inspection to determine compliance of dies, collets, and diestocks with this specification. Results of this inspection shall be considered final and acceptance or rejection shall be based thereon.

4.5 Inspection of preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-H-15424. Levels of preservation and packaging and level of packing shall be as specified (see 6.2).

5.2 Marking.

5.2.1 Military agencies. In addition to any special marking required by the contract, or order, or herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Types I and II dies are adjustable threading tools used with diestocks for cutting loose, normal, or tight external threads on screws, bolts, or other male parts.

6.1.2 Type III rethreading dies are used for repair work, rethreading damaged or rusted external threads, but may be used occasionally for cutting new threads. They require no special holder or stock. Any wrench large enough to fit will suffice.

6.1.3 Type IV collets are used to hold present 5/8-inch outside diameter type I dies to specific pitch diameters.

6.1.4 Types V and VI diestocks are used with screw and bolt threading dies.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification
b. Type and series of dies, collets, and diestocks required (see 1.2).
c. If dies are to be left-hand thread (see 3.5.1.1.1).
d. Dimensions and threads per inch or pitch (if metric) required for dies (see tables I, II, III, or IV).
e. Dimensions required for diestocks (see tables V or VI).
f. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Pipe dies. Pipe threading dies and diestocks are covered by GGG-T-581.

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